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The impact cratering on Titan: The history of impact melt production

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Abstract

Impact craters on Titan provide possible habitable environments. Organic material from the impactor body, along with the organics on Titan's surface, mix with the melt, making way for possible aqueous prebiotic chemistry to occur in these melt-pool environments. There is the possibility that impactors can fracture the ice or even reach the subsurface ocean directly, allowing the transport of organic material to the subsurface ocean, providing another possible environment for prebiotic chemistry. In this work, we use a model for the impact rate on Titan to estimate the total number of impacts to be 339. This produces a total volume of melt of $8.98 \times 10^6 \text{ km}^3$ and $8.75 \times 10^5 \text{ km}^3$ for a pure water ice surface and methane clathrate crust respectively. We discuss the implications for prebiotic chemistry in these melt pool environments and in the subsurface ocean.

1 Introduction

Titan – Saturn's largest moon – is a compelling candidate for habitability. Titan has a dense atmosphere, primarily composed of nitrogen and methane, and has an organic-rich layer on its surface. Its interior is surrounded by a 40 to 170 *km* thick ice shell, likely capped with a methane clathrate crust layer of 5 to 15 *km* [1] [2]. Titan is also suggested to host a subsurface ocean beneath its thick ice shell. Impactor bodies able to hit Titan's surface melt the ice, creating liquid melt pools. This enables the mixing of organics from Titan's surface and the impactor body within liquid water environments. This may provide the possibility for aqueous prebiotic chemistry.

If the impactor body is large enough, this can allow the transport of organics to the subsurface ocean. This could occur by direct breaching of the ice-ocean interface – Menrva (the largest impactor on Titan) has been suggested to have done this [3]. Smaller impactors not able to reach the subsurface ocean directly, may fracture the ice, enabling the organics to reach the ocean indirectly. Carnahan et al., (2022) says that over 40% of melt is transported to the subsurface ocean, if the crater depth exceeds half of the ice shell thickness [4]. The mixing of organic material within the subsurface ocean is another environment potentially able to undergo prebiotic chemistry.

Most impacts on Titan come from the Primordial Kuiper Belt (PKB), initially ~ 30 Earth masses located between ~ 24 and $\sim 30 \text{ au}$ [5]. The cometary objects in the PKB, composed of ice and rock, vary in size and density depending on their porosity of ice. When Neptune migrated outward across the PKB – around 20 *Myr* after the solar nebula dispersed – this put a lot of PKB objects onto destabilised orbits, with many crossing the orbits of the giant planets [5]. This destabilised population is the most probable source of most early impacts on Titan. As Neptune reached approximately 28 *au*, this caused the giant planets to break out of their resonant chain, causing them to migrate – Jupiter migrating inwards, and the others migrating outwards. This giant planet instability prompted even more collisions between the destabilised population and the giant planets and their satellites. The impact rate on Titan was at its highest during this period. The destabilised population then massively decreased due to collisions with other bodies, but most being ejected out of the Solar System. The objects that survived make up the scattered disk, which is the source of the most recent impacts on Titan. By this point the impact rate on Titan had decreased by several orders of magnitude [6].

Titan is thought to have a methane clathrate crust layer on top of its ice shell. This provides insulation, reducing the extraction of heat from the ice shell and subsurface ocean below it, which lowers the ocean's freezing rate [7]. Kalousová and Sotin (2020) say that a 7.5 to 15 *km* thick methane clathrate layer should be present to explain the lack of crystallisation or freezing of the subsurface ocean [7]. However, the impact craters have a strangely shallow depth, implying a less thick methane clathrate layer is present, due to its higher density [8]. Simulations show that a 5 *km* methane clathrate thickness best reproduces the observed crater topography [8]. Although much previous work suggests evidence of a methane clathrate crust layer, we have no direct evidence of its existence [8]. Therefore, in this work we compare a (non-porous) 5 *km* methane clathrate crust layer with a pure water ice surface of thickness 100 *km* (with 20% porosity). We discuss how these two surfaces vary the crater size and melt production due to their different densities [2].

Previous work on the impact cratering on Titan assume a constant impact rate or a simple $1/t$ dependence [9] [10]. In this work, we use a more accurate impact rate on Titan from the model in Bottke et al., (2024), which is based on simulations in Nesvorný et al., (2019) [11]. We, therefore, determine how much melt is produced by impact craters at different points in Titan's history.

2 Methods

2.1 Impact rate on Titan

To plot the impact rate on Titan, we firstly plot the impact rate on Jupiter because of its significant influence on the impacts on other planets and satellites. Bottke et al., (2024) models the impact rate on Jupiter over its 4.5 Gyr lifetime from the destabilised population and scattered disk, based on numerical simulations described in Nesvorný et al (2019) [6] [11]. In Figure [1], we estimate Bottke's plot of Jupiter's impact rate as a series of negative power laws, which demonstrates the overall decrease in impacts over time. Approximately 1.1% of PKB objects impact onto Jupiter over its lifetime, and by 40 Myr after Neptune entered the PKB approximately 90% of impacts have occurred [6].

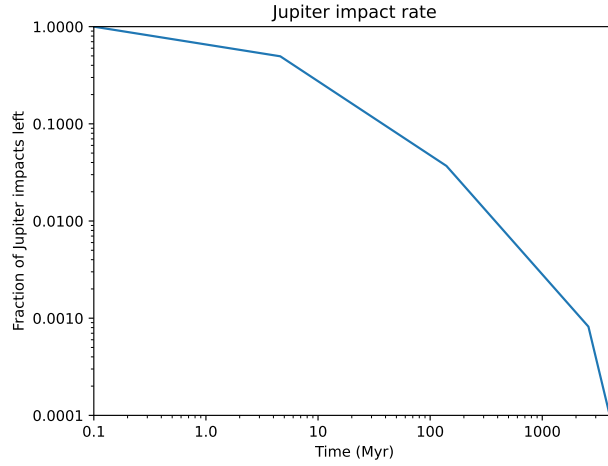


Figure 1: Fraction of impacts left to hit Jupiter from the destabilised population and scattered disk over 4.5 Gyr. The initial time is when Neptune enters the PKB.

Assuming the initial PKB is ~ 30 Earth masses and the PKB objects have an average density of 0.6 g cm^{-3} and an average diameter of 100 km , we obtain an estimate for the total number of impacts on Jupiter [5] [12]. Multiplying by the probability factor, 5.4×10^{-5} , of a comet hitting Titan relative to hitting Jupiter, we estimate the total number of impacts on Titan to be approximately 339. [9]. Figure [2] shows the impact rate on Titan, illustrating that most of the 339 impacts hit Titan within 4.5 Gyr after Neptune entered the PKB.

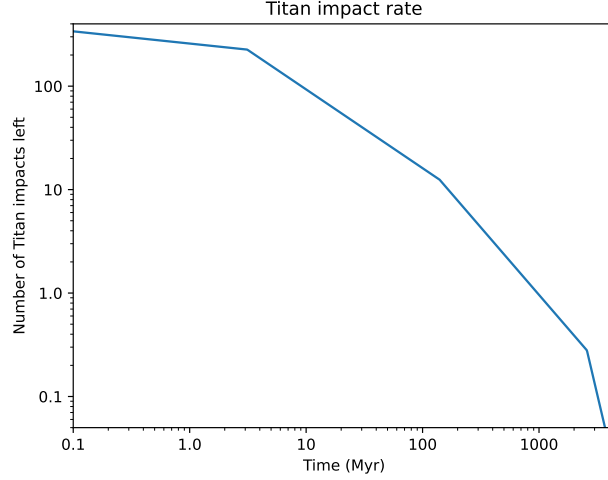


Figure 2: Total number of impacts left to hit Titan from the destabilised population and scattered disk over 4.5 Gyr. The initial time is when Neptune enters the PKB.

The PKB objects vary in size from small dust particles to large planetesimals. The size frequency distribution of impactors on Titan is deduced from craters on the Galilean satellites [9]. We use the scaling law described in Korycansky and Zahnle (2005), assuming an impact rate of 4.5×10^{-5} on Titan (20% lower than Jupiter's impact rate). [9] [13]. Figure [3] shows the size frequency distribution of impactors on Titan.

Many comets that hit Titan undergo significant deformation or ablation as they pass through Titan's dense atmosphere. This effects their impact velocity, diameter, and for very small, low density comets, get destroyed entirely before they can reach the surface [14]. Anslow et al., (2025) derives the minimum radius of the impactor body, $r_{\min}$, that avoids any fragmentation (complete destruction): [14]

$$r_{\min} \approx 100 \left(\frac{P_{\text{surf}}}{1 \text{ bar}} \right) \left(\frac{\rho_{\text{imp}}}{0.4 \text{ g cm}^{-3}} \right)^{-1} \left(\frac{g}{9.81 \text{ m s}^{-2}} \right)^{-1} \left(\frac{\sin \theta}{1/\sqrt{2}} \right)^{-1} \text{ m} \quad (1)$$

where P_{surf} is the atmospheric surface pressure, ρ_{imp} is the impactor density, g is the surface gravity and θ is the impact angle perpendicular to the surface.

Titan's atmospheric surface pressure is 1.5 bar and its surface gravity is 135 cm s^{-2} [10]. Assuming an impact angle of 45° and impact velocity of 11.2 km s^{-1} , the minimum impactor diameter able to hit Titan's surface avoiding fragmentation is $\sim 0.727 \text{ km}$ [6]. In this work, we assume that bodies smaller than 1 km in diameter are unable to reach the surface due to complete fragmentation in the atmosphere [10].

Larger bodies may undergo ablation, affecting their diameter and other properties, which is described by the pancake model in Anslow et al., (2025) [14]. Neish et al., (2024) says that comets smaller than 20 km in diameter that hit Titan will undergo ablation [2]. However, ablation only involves a fractional loss of the body's mass, so for simplicity we will assume ablation is negligible [14].

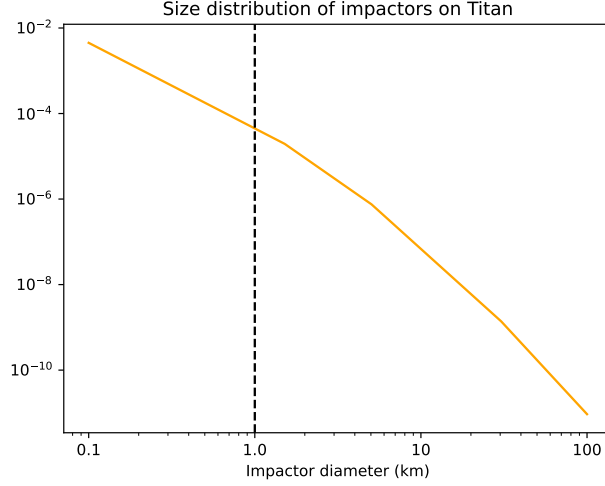


Figure 3: The size frequency distribution of impactor bodies that hit Titan from the destabilisation population and scattered disk. The impactor diameters are evaluated in the following bins: $0.1 - 1.5 \text{ km}$, $1.5 - 5 \text{ km}$, $5 - 30 \text{ km}$, $30 - 100 \text{ km}$. The power laws for each size bin are described in Korycansky and Zahnle (2005).

2.2 Impact crater sizes

For the transient crater diameter D_t (immediate crater diameter after impact), we use the scaling law from Schmidt and Housen (1987): [15] [12]

$$D_t = 1.161 \left(\frac{\rho_{\text{imp}}}{\rho_{\text{tar}}} \right)^{1/3} D_{\text{imp}}^{0.78} v_{\text{imp}}^{0.44} g^{-0.22} \sin^{1/3}(\theta) \quad (2)$$

where D_{imp} is the impactor diameter, ρ_{imp} is the impactor density, ρ_{tar} is the target density, v_{imp} is the impactor velocity, θ is the impact angle perpendicular to the surface, and g is the acceleration due to gravity on Titan.

We compare how a target density of 0.93 g cm^{-3} (pure water ice) varies with a target density of 0.95 g cm^{-3} (5 km methane clathrate crust) [8] [16]. Figure [4] shows how the transient crater diameter varies with the impactor diameter for the two surfaces. There is a slight difference between a pure water ice surface and a 5 km methane clathrate crust, where the former produces a larger transient crater. The difference becomes more prominent for larger impactors.

Very large impactor bodies will not leave a crater, but cause catastrophic dispersion to the target body. The catastrophic threshold is defined as the minimum impactor size to cause the target body to lose half its mass [17]. The specific energy of the impactor required to cause such an event is described in Housen and Holsapple (1990), which relates to the impactor's radius and collision speed [18]. To obtain the minimum impactor size that will cause catastrophic dispersion to Titan, we work in the gravity regime, because its radius is larger than 20 km , so its gravitational effects have a significant influence on the catastrophic threshold [17]. Housen and Holsapple (1990) derive a relationship between the maximum transient crater radius R_c^* and the target body's radius R :

$$\frac{R_c^*}{R} = \text{constant} \quad (3)$$

where the constant value is 0.32, which was obtained experimentally [18]. Thus, we get a maximum crater size of 824 km for Titan. This corresponds to an impactor diameter of 231 km or 234 km for a pure water ice or methane clathrate surface respectively. Impactors larger than this will cause catastrophic dispersion, hence I will assume a maximum impactor diameter of 231 km .

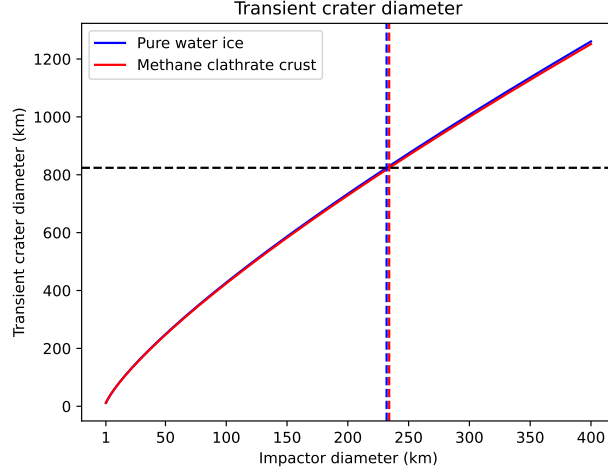


Figure 4: The transient crater diameter over the impactor diameter, with initial impactor diameter of 1 *km*. The maximum transient crater diameter at the catastrophic threshold is 824 *km* for Titan. The corresponding impactor diameters at this threshold is 231 *km* and 234 *km* for a pure water ice surface and methane clathrate surface respectively.

Transient craters undergo enlargement after their formation. Small transient craters expand due to the collapse of their steep rim walls which slide inwards, resulting in simple craters [19] [6]. Large transient craters collapse into much wider and shallower craters, resulting in complex craters. The transition size between simple and complex craters was estimated experimentally by Aponte-Hernández et al., (2021), which shows an inverse scaling with the surface gravity g : [20]

$$D_{sc} = 39.7 g^{-0.4} \quad (4)$$

The scaling laws for simple and complex craters are described in Anslow et al., (2025), following from Croft (1985): [12] [21]

$$\begin{aligned} D_{\text{simp}} &= 1.25 D_t \\ D_{\text{comp}} &= 1.17 D_t^{1.13} D_{sc}^{-0.13} \end{aligned} \quad (5)$$

For Titan the simple-complex transition crater diameter D_{sc} is approximately 5.6 *km*, which corresponds to a small impactor diameter of 1 *km* from Figure [4]. Hence we assume all impact craters on Titan are complex. Figure [5] illustrates how the final crater diameter varies with the impactor diameter, showing that a pure water ice surface produces a slightly larger crater diameter – analogous to the results in Figure [4].

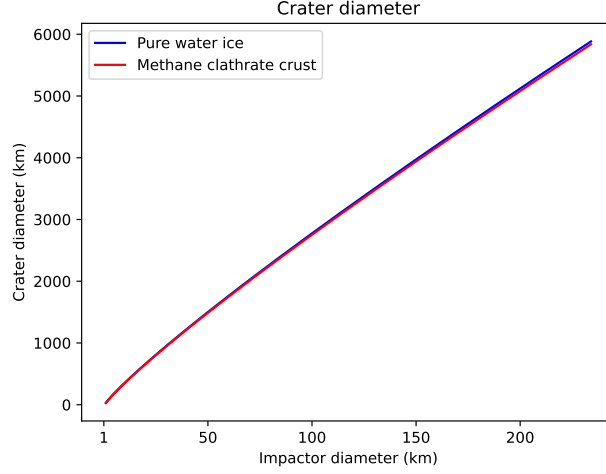


Figure 5: The final crater diameter after collapse for varying impactor diameters. The initial impactor diameter is 1 km . Complex craters form from impactor diameters ≥ 1 km , so all crater diameters are assumed to be complex.

We now determine the amount of melt produced within the impact craters using the scaling law described in Neish et al., (2024) [2]. The volume of melt produced for impact craters onto a pure water ice surface:

$$V_{ice} = 0.01 R_c^{3.27} \quad (6)$$

for a methane clathrate crust:

$$V_{clath} = 0.001 R_c^{3.27} \quad (7)$$

where V is the volume of melt produced and R_c is the crater radius.

Figure [6] demonstrates that an impactor body hitting a pure water ice surface produces significantly more melt than hitting a methane clathrate crust. This is due to the scaling difference of a factor 10, and is explained by the higher porosity of pure water ice over methane clathrate.

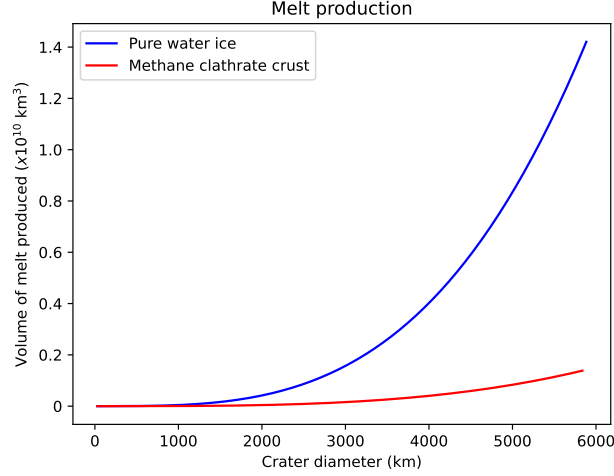


Figure 6: The volume of melt produced for different crater diameters for a pure water ice surface and methane clathrate crust. Significantly more melt is produced when the surface is pure water ice due to its higher porosity and lower density.

To determine which impactor bodies are able to reach the subsurface ocean, we calculate the transient crater depth. This depends on the Froude number Fr and Mach number M , which are defined as follows: [22]

$$Fr = \frac{v^2}{gR} \quad (8)$$

$$M = \frac{v}{U_s} \quad (9)$$

where v is the impactor velocity, R is the crater radius, U_s is the speed of sound in the target material and g is the surface gravity.

Assuming the speed of sound is 4 km s^{-1} and 3 km s^{-1} in pure water ice and methane clathrate respectively, $M \geq 1$ for both materials (the impactor velocity is faster than the speed of sound in both target materials), so the impacts on Titan are considered to be supersonic collisions [22].

We use the scaling law for the transient crater depth Z_c from Allibert et al., (2023): [22] [23]

$$\frac{Z_c}{R Fr^{1/4}} = a (1 + bM^2)^{-c} \quad (10)$$

where R is the crater radius and the coefficients $a = 1.092$, $b = 0.11$, $c = 0.25$, which were obtained numerically [22]. These coefficients apply to both pure water ice and methane clathrate surfaces since simulations show they do not vary significantly for different materials [22]. Substituting in the equation for Fr and re-arranging, one can obtain a relationship between the transient crater depth Z_c and the impactor diameter D :

$$Z_c = \frac{(D/2)^{3/4} v^{1/2}}{g^{1/4}} a (1 + bM^2)^{-c} \quad (11)$$

Figure [7] shows how the transient crater depth varies with the impactor diameter. A pure water ice surface produces larger crater depths because of its lower density than methane clathrate. For an impactor body

to reach the subsurface ocean directly, it must produce a crater depth of at least 100 *km* for a pure water ice surface or 105 *km* for a methane clathrate crust. This corresponds to a minimum impactor diameter of 22.4 *km* for a pure water ice surface or 26.5 *km* for a methane clathrate crust.

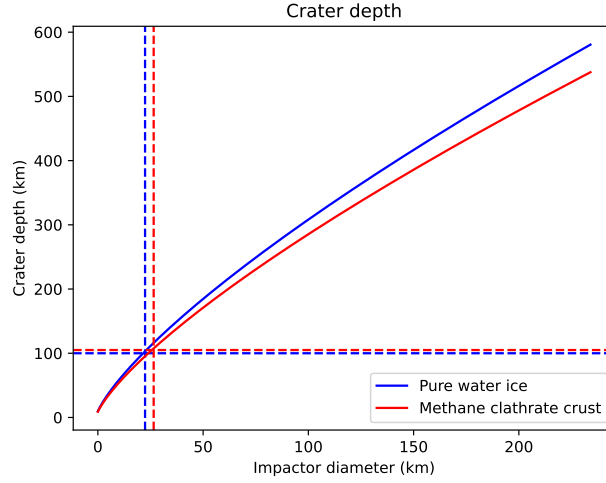


Figure 7: The transient crater depth over the impactor diameter for a pure water ice surface and methane clathrate crust. For an impactor body to reach the subsurface ocean directly, the resulting crater depth must be at least 100 *km* for a pure water ice surface or at least 105 *km* for a methane clathrate crust. This corresponds to a minimum impactor diameter of 22.4 *km* for a pure water ice surface or 26.5 *km* for a methane clathrate crust.

3 Results

3.1 Total melt production

From Figure [2] Titan's impact rate can be evaluated in four time periods: $\Delta t_0 = 0.1 - 3.12 \text{ Myr}$, $\Delta t_1 = 3.12 - 140 \text{ Myr}$, $\Delta t_2 = 140 - 2578 \text{ Myr}$, $\Delta t_3 = 2578 - 4500 \text{ Myr}$. There is an overall decrease in impacts since Neptune entered the PKB, but the impact rate is at its highest in Δt_1 . This is likely due to the giant planet instability event which occurred in this time period; either 10.5 or 32.5 *Myr* after Neptune entered the PKB [5]. The total number of impacts on Titan is estimated to be 339. From Figure [2] we determine the number of impacts in each time period as follows: $\Delta t_0 = 113$, $\Delta t_1 = 213$, $\Delta t_2 = 12$, $\Delta t_3 = 1$.

In Figure [3] we used the scaling law by Korycansky and Zahnle (2005) to obtain the size distribution of impactors onto Titan [9]. This is evaluated in four size ranges, but here we evaluate in five size ranges to separate the impactors $< 1 \text{ km}$ which do not hit Titan's surface due to fragmentation [9] [14]. The size ranges are as follows, where D is the impactor diameter: $\Delta D_0 = 0.1 - 1 \text{ km}$, $\Delta D_1 = 1 - 1.5 \text{ km}$, $\Delta D_2 = 1.5 - 5 \text{ km}$, $\Delta D_3 = 5 - 30 \text{ km}$, $\Delta D_4 = 30 - 100 \text{ km}$. From Figure [3] we estimate the number of impacts in each size range: $\Delta D_0 = 314$ (93%), $\Delta D_1 = 12$ (3.4%), $\Delta D_2 = 12$ (3.5%), $\Delta D_3 = 1$ (0.35%), $\Delta D_4 = 0$ (0.000031%). This tells us that the majority of impactors onto Titan (93%) are $< 1 \text{ km}$ in diameter, which are not able to hit the surface because they are destroyed in the atmosphere. Approximately 7% of impactors actually hit Titan's surface, so only 24 out of the total 339 impactors hit the surface, producing a crater.

To determine the total amount of melt produced by these impactors, we first calculate the average crater diameter $\bar{D}_c$ and average volume of melt produced $\bar{V}_{melt}$ for each impactor diameter range, using Figures [5] and [6]. This is demonstrated in Table [1] for a pure water ice surface and methane clathrate crust.

ΔD (km)	$\bar{D}_c$ (km)		$\bar{V}_{melt}$ (km ³)	
$\Delta D_1 = 1 - 1.5$	45.3	44.9	2.00×10^3	1.94×10^2
$\Delta D_2 = 1.5 - 5$	104	103	4.92×10^4	4.80×10^3
$\Delta D_3 = 5 - 30$	451	447	8.37×10^6	8.15×10^5
$\Delta D_4 = 30 - 100$	1450	1440	2.76×10^8	2.69×10^7

Table 1: Average crater diameter $\bar{D}_c$ and average volume of melt produced $\bar{V}_{melt}$ for each impactor diameter range ΔD . LHS values for a pure water ice surface and RHS values for a methane clathrate crust.

We can see that significantly more melt is produced for a pure water ice surface (about an order of magnitude higher) even though the crater diameters between the two surfaces are similar. This is also evident in Figures [5] and [6].

Using the number of impacts in each size range that hit Titan's surface ($\Delta D_1 = 12$, $\Delta D_2 = 12$, $\Delta D_3 = 1$, $\Delta D_4 = 0$), we obtain the total volume of melt produced over Titan's lifetime of 4.5 Gyr to be 8.98×10^6 km³ and 8.75×10^5 km³ for a pure water ice surface and methane clathrate crust respectively.

In Table [2] we determine the volume of melt produced in each time period. We use the number of impacts in each time period: $\Delta t_0 = 113$, $\Delta t_1 = 213$, $\Delta t_2 = 12$, $\Delta t_3 = 1$, and the approximation above that 3.4% of impactors lie in ΔD_1 , 3.5% lie in ΔD_2 , 0.35% lie in ΔD_3 , and $\sim 0\%$ lie in ΔD_4 .

Δt (Myr)	$\bar{V}_{melt,ice}$ (km ³)	$\bar{V}_{melt,clath}$ (km ³)
$\Delta t_0 = 0.1 - 3.12$	2.03×10^5	1.98×10^4
$\Delta t_1 = 3.12 - 140$	8.73×10^6	8.50×10^5
$\Delta t_2 = 140 - 2578$	4.92×10^4	4.80×10^3
$\Delta t_3 = 2578 - 4500$	0	0
Δt_{total}	8.98×10^6	8.75×10^5

Table 2: The average volume of melt produced for a pure water ice surface $\bar{V}_{melt,ice}$ and methane clathrate crust $\bar{V}_{melt,clath}$ in each time period Δt , and the total volume of melt produced over Titan's lifetime of 4.5 Gyr.

We see that most melt produced is within Δt_1 because the majority of impacts happen during this time period. No melt is produced in Δt_3 because we expect approximately one impactor during this time period, and this is likely to be a small impactor which will be fragmented in the atmosphere, hence will not hit the surface.

3.2 Total melt transported to subsurface ocean

If an impactor body produces a large enough crater it can breach the ice-ocean interface. So, all the melt produced will be transported to the subsurface ocean. In Figure [7] we saw that a minimum impactor diameter of 22.4 km or 26.5 km for a pure water ice surface or methane clathrate crust respectively is required to breach the ice-ocean interface. This encompasses all impactors within the ΔD_4 range and the impactors at the upper end of the ΔD_3 range. However, we previously determined that there will unlikely be impactors within the ΔD_4 range, but possibly one impactor in the ΔD_3 range.

Let us assume one impactor above 22.4 km or 26.5 km for a pure water ice surface or methane clathrate crust respectively does hit Titan's surface. Then the resulting volume of melt transported to the subsurface ocean will be 8.37×10^6 km³ or 8.15×10^5 km³ for a pure water ice surface or methane clathrate crust respectively, as shown in Table [1]. So, all the melt produced by these impactors will be transported to the subsurface ocean.

The transport of melt to the subsurface ocean can also be possible for smaller impactors due to the fracturing of ice. Carnahan et al., (2022) states that if the crater depth exceeds half the ice shell thickness,

over 40% of melt will be transported to the subsurface ocean [4]. In this work, we assume 65% of melt is transported to the subsurface ocean if the crater depth reaches half the ice shell thickness, which is the average calculated in Carnahan et al., (2022) [4] [2].

To reach half the ice shell thickness of 100 km thick, a crater depth of 50 km is required. If we have a 105 km thick ice shell due to the presence of a methane clathrate crust, a crater depth of 52.5 km is required. From Figure [7] the corresponding impactor diameters required to reach these depths are 8.9 km and 10.5 km respectively. These lie in the ΔD_3 range, but we only expect one impactor in this size range to hit Titan.

If we assume one impactor at least 8.9 km or 10.5 km in diameter hits Titan for a pure water ice surface or methane clathrate crust respectively, we expect approximately $5.44 \times 10^6 \text{ km}^3$ or $5.30 \times 10^5 \text{ km}^3$ of melt to be transported to the subsurface ocean.

We expect one impactor in the ΔD_3 size range to hit Titan's surface. The volume of melt transported to the subsurface ocean for this impactor will vary between just less than 65% to all the melt produced being transported. So a maximum of $8.37 \times 10^6 \text{ km}^3$ or $8.15 \times 10^5 \text{ km}^3$ will be transported and the minimum lies just less than $5.44 \times 10^6 \text{ km}^3$ or $5.30 \times 10^5 \text{ km}^3$ for a pure water ice surface or methane clathrate crust respectively. Smaller impactors in the ΔD_1 and ΔD_2 size ranges may also transport a small proportion to the ocean. We assume that in addition to this the above value is acceptable to take as the minimum. We now take the average of melt transported to the subsurface ocean of both ranges. We obtain $6.91 \times 10^6 \text{ km}^3$ and $6.73 \times 10^5 \text{ km}^3$ for a pure water ice surface and methane clathrate crust respectively.

Due to some melt reaching the subsurface ocean, this will reduce the amount of melt in the crater pools at the surface. We calculated above that the total volume of melt produced is $8.98 \times 10^6 \text{ km}^3$ and $8.75 \times 10^5 \text{ km}^3$ for a pure water ice surface and methane clathrate crust respectively. Therefore, the volume of melt left in the crater pools is $2.07 \times 10^6 \text{ km}^3$ and $2.02 \times 10^5 \text{ km}^3$ respectively. So, approximately 23% of melt is transported to the subsurface ocean.

4 Discussion

4.1 Uncertainty of Titan's impact rate

Modelling the impact rate on Titan, or any planetary satellite, is a challenge due to uncertainties regarding its formation. In this work, we assume Titan's age is 4.5Gyr, the age of the solar system, however many suggest Titan is a young satellite. Titan may have formed from the circumplanetary disk of dust and gas around Saturn [24], or may have formed from a massive ice-rich ring close to Saturn's Roche limit [25]. Models suggest Titan's age may be as young as 2800 Myr [6]. Uncertainty with Titan's age makes it difficult to determine the number of impacts onto Titan. The value of 339 approximated in this work may be too large compared to the actual number of impacts. It is also difficult to constrain this value based on Titan's observed crater topography. This is because Titan has a very geologically active surface, so many craters disappear over time due to the replenishing of surface material. The observed crater topography, therefore only provides a lower bound to the total number of impacts.

To approximate the total number of PKB objects in order to obtain total number of impacts on Titan, we assumed an average comet density of 0.6 g cm^{-3} and average comet diameter of 100km [12] [5]. But comet densities vary, for example less porous comets can be upto 1 g cm^{-3} . However, comet densities are well constrained to $0.4 - 0.7 \text{ g cm}^{-3}$, so an assumption of 0.6 g cm^{-3} is reasonable [12]. Assuming an average comet diameter of 100km in the PKB is less of a reasonable assumption, because these objects encompass a wide range of sizes ranging from small dust particles to large planetesimals [5] [9]. These assumptions add further uncertainty to the total number of impacts on Titan determined in this work, and thus the impact rate on Titan.

4.2 Survival of organics

The possible synthesis of prebiotic feedstock molecules within melt pool environments was first suggested by Thompson and Sagan (1992) [26]. Organic molecules, such as hydrocarbons and nitriles, present on the comet and Titan's surface mix in the liquid melt providing the opportunity for aqueous prebiotic chemistry [27]. This can involve the production of the essential amino acids for life.

However, during atmospheric entry cometary bodies will experience volatile loss. Organics will be stripped off the body, reducing the amount of organics left to reach the surface [27]. As well as this, the surface is heavily shocked at impact, so many of the organic molecules are destroyed and ejected out of the crater region [28]. Todd and Öberg (2020) models the survival of hydrogen cyanide during impacts, and finds a larger impactor size, larger impact velocity and smaller impact angle will decrease the chances of its survival [29] [27]. However, the surface located behind the impact trajectory is only lightly shocked, so many of these organic molecules survive and are deposited in the liquid melt pool in the crater [28].

4.3 Freezing of melt pools

The total volume of melt produced over 4.5 Gyr of Titan's lifetime does not represent how much melt is on Titan at present. Titan's cold environment means the impact melt pools will eventually freeze over. In this work we do not account for the freezing timescale of these melt pools, but this presents a significant error with our calculations. O'Brien et al., (2005) states that craters 15km in diameter will freeze completely after approximately $10^2 - 10^3$ years, and 150km crater diameters will last for $10^3 - 10^4$ years [30]. Mostly all craters on Titan are larger than 15km in diameter from Figure [5], so 10^2 years can be considered a lower bound of the freezing timescale for melt pools on Titan.

The freezing of these melt pool environments will limit the possibility of prebiotic chemistry occurring in these environments. Smaller impactors produce less melt than larger impactors, so these smaller melt pools will freeze quicker, which may not allow for prebiotic chemistry to take place.

The freezing of melt pools drastically influences how much melt is transported to the subsurface ocean. Carnahan et al., (2022) says that the drainage of melt to the subsurface ocean occurs on timescales of $10^3 - 10^4$ years [4]. So, in many cases the freezing rate is quicker than the drainage of melt to the ocean. Therefore, if the crater depth reaches half of the ice shell thickness, the assumption that 65% of melt will be transported to the ocean may not hold [4] [2]. The actual amount of melt transported may be much less.

In this work, we do not consider the amount of melt lost in the craters due to the transport to the subsurface ocean. This will reduce the amount of organic molecules left in the melt pools, which may drastically limit the possibility for prebiotic chemistry.

To determine the freezing timescale of these melt pool environments, it is crucial to know the types of organic molecules present and in what quantities. This is because each organic molecule will have its own freezing timescale, therefore affecting the overall freezing rate. Niesh et al., (2024) outlines the fraction of hydrocarbons and nitriles likely present in these melt pools [2]. Future work on the freezing rate of these melt pools will better constrain the amount of melt transported to the subsurface ocean and help us better understand whether prebiotic chemistry can occur in the melt pool environments and subsurface ocean.

4.4 Presence of subsurface ocean

Thermal-evolution models suggest the existence of a subsurface ocean beneath the thick ice sheet inferred from tidal-heating on Titan [1]. Tidal-heating is due the strong gravitational pull between Titan and Saturn causing deformation in Titan's icy body. However, Petricca et al., (2025) say that the tidal-heating of Titan is too large and is inconsistent with the presence of a subsurface ocean [31]. This is because a liquid ocean would dampen the tidal-heating, so a lower tidal-heating value would be expected. Petricca et al., (2025) suggest that there instead exists a slushy high-pressure ice layer below the thick ice sheet [31].

In this work, we discuss the implications of prebiotic chemistry within the subsurface ocean due to the transport of organics from the melt pool environments. If Titan does not in fact host a subsurface ocean,

much of this work regarding to the subsurface ocean is not applicable. So, only the implications of prebiotic chemistry within the melt pool environments is relevant.

5 Conclusion

In this work, we determined the amount of melt produced by impact craters on Titan within different time periods and over its 4.5 Gyr lifetime. We used an impact rate model for Titan based on Jupiter's impact rate and simulations in Nesvorný et al., (2019) [11] [5]. We determined the size distribution of impactors onto Titan and found that impactors $> 1\text{ km}$ in diameter are able to pass through Titan's dense atmosphere without fragmentation, and hence are able to hit the surface [14].

We considered two possible surfaces for Titan: a pure water ice surface of thickness 100 km ; and a methane clathrate crust of 5 km on top of this ice shell. We modelled the transient and final crater sizes and compared how they differ based on the type of surface. We indicated that the maximum transient crater diameter on Titan is 824 km , so an impactor diameter larger than 231 km or 234 km for a pure water ice surface or methane clathrate crust respectively will cause catastrophic dispersion to Titan. From the crater sizes we determined the amount of melt produced within the crater basins and how much melt is transported to the subsurface ocean.

We found that the approximate number of impacts that hit Titan is 339, producing a total of 8.98×10^6 and 8.75×10^5 for a pure water ice surface and methane clathrate crust respectively. We also determined the amount of melt produced in four time periods, which showed that most impactors on Titan occurred in Δt_1 , likely due to the giant planet instability event.

We also determined the amount of melt transported to the subsurface ocean to be approximately 5.44×10^6 or 5.3×10^5 for a pure water ice surface or methane clathrate crust respectively. This is assuming that 65% of melt is transported to the ocean for a crater depth that exceeds half the ice shell thickness [2] [4].

These calculated melt volumes can be constrained better when considering the freezing timescales of the melt pools, which depends on the types of organic molecules present and the crater size.

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